Regularized Regression for House Price Prediction: A Comparative Study of OLS, Ridge, LASSO, and Elastic Net on the Ames Housing Dataset

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Abstract

This paper compares four linear regression approaches – Ordinary Least Squares (OLS), Ridge regression, the Least Absolute Shrinkage and Selection Operator (LASSO), and the Elastic Net – for predicting residential sale prices in the Ames, Iowa housing market. Using a cleaned subset of $n = 2{,}928$ observations and fourteen predictors covering structural, qualitative, and locational characteristics of each property, we fit a log-linear specification and evaluate the four estimators both in-sample, via coefficient significance and shrinkage behavior, and out-of-sample, via a held-out test-set root mean squared error (RMSE). All four models explain approximately 88% of the variance in log sale price and achieve nearly identical predictive accuracy (test RMSE between 0.1637 and 0.1645 on the log scale). We find that regularization in this setting contributes primarily to model parsimony and interpretability – LASSO and the Elastic Net eliminate several weak neighborhood indicator variables and the redundant room-count predictor – rather than to a measurable improvement in predictive accuracy, a result consistent with the sample size being large relative to the number of predictors.

Keywords: hedonic pricing, regularization, LASSO, Ridge regression, Elastic Net, cross-validation, real estate valuation

1 Introduction

The prediction of residential sale prices is a classical application of hedonic pricing theory, in which the price of a differentiated good – a house – is modeled as a function of its constituent characteristics. Beyond its practical relevance for appraisal and mass-valuation systems, the problem is a useful testbed for comparing linear modeling strategies, since housing datasets typically combine a moderate number of continuous, ordinal, and categorical predictors with well-documented multicollinearity (e.g., overall living area and room counts, or basement area and total square footage).

This paper uses the Ames Housing dataset [1], a widely used alternative to the Boston Housing dataset for regression benchmarking, to compare four estimators of increasing regularization: OLS, which places no penalty on coefficient size; Ridge regression [3], which imposes an ℓ_2 penalty and shrinks coefficients smoothly toward zero without eliminating them; LASSO [4], which imposes an ℓ_1 penalty and can shrink coefficients exactly to zero, performing implicit variable selection; and the Elastic Net [5], which combines both penalties through a mixing parameter α.

The remainder of the paper is organized as follows. Section 2 describes the data and preprocessing steps. Section 3 presents the model specification and estimation procedure for each method. Section 4 reports in-sample and out-of-sample results. Section 5 discusses the comparative findings, and Section 6 concludes.

2 Data

2.1 Source and Variable Selection

The dataset comprises 2,930 individual residential property sales recorded in Ames, Iowa. From the full set of available columns, fourteen variables were retained for the analysis: the response variable `SalePrice`, and thirteen predictors spanning structural characteristics (`Gr.Liv.Area`, `Total.Bsmt.SF`, `Garage.Cars`, `Year.Built`, `Full.Bath`, `TotRms.AbvGrd`, `Lot.Area`), qualitative assessments (`Overall.Qual`, `Overall.Cond`, `Kitchen.Qual`), and categorical descriptors (`Neighborhood`, `Bldg.Type`, `Central.Air`).

Several additional columns in the raw dataset (e.g., `Pool.QC`, `Alley`, `Fence`, `Fireplace.Qu`) exhibit a high proportion of missing values. In these cases, missingness is structural rather than incidental: an `NA` entry indicates the absence of the feature (no pool, no alley access, no fence) rather than an unrecorded value. These variables were deliberately excluded to avoid a separate imputation or recoding stage, and the analysis retains only variables for which missingness is negligible and attributable to data-entry gaps.

2.2 Missing Data and Cleaning

Among the fourteen selected variables, only `Total.Bsmt.SF` and `Garage.Cars` contain a single missing observation each, corresponding to an overall missingness rate of 0.0049%. The two incomplete rows were removed via listwise deletion, reducing the working sample from 2,930 to $n = 2{,}928$ observations.

2.3 Response Transformation

The raw `SalePrice` variable is strongly right-skewed as a result of a subset of high-value properties. To better satisfy the homoscedasticity and approximate-normality assumptions underlying linear regression, the response variable was log-transformed:

$$y = \log(\text{SalePrice}). \tag{1}$$

Figure 1 illustrates the effect of the transformation: the distribution of `SalePrice` is markedly right-skewed, while $\log(\text{SalePrice})$ is close to symmetric.

2.4 Categorical Encoding

The variables `Neighborhood`, `Bldg.Type`, `Kitchen.Qual`, and `Central.Air` were converted to factors. `Neighborhood` alone generates 27 dummy variables upon expansion, so that the full design matrix used for the regularized models contains approximately 40 predictors in total – providing LASSO and the Elastic Net with a realistic opportunity to perform genuine variable selection, as opposed to a setting with only a handful of numeric predictors.

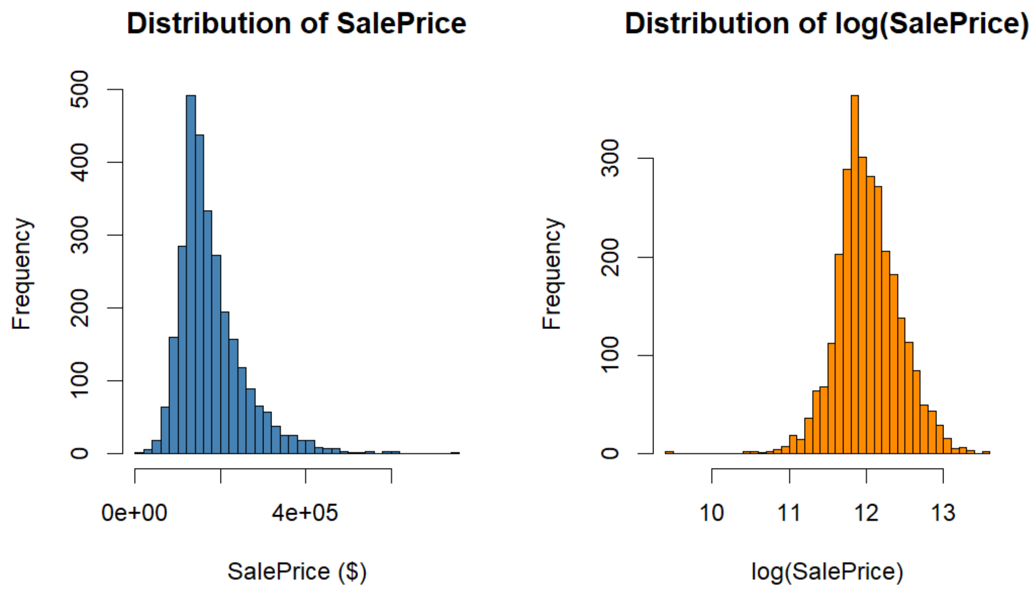

Figure 1: Distribution of raw sale price (left) versus log sale price (right). The log transformation substantially reduces right-skewness.

3 Methodology

3.1 Model Specification

All four models share the same linear specification for the log sale price:

$$y_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij} + \varepsilon_i, \qquad i = 1, \ldots, n, \tag{2}$$

where y_i is the log sale price of property i, x_{ij} are the p predictors (numeric variables and dummy-encoded categorical levels), and ε_i is an idiosyncratic error term.

3.2 Ordinary Least Squares

The OLS estimator minimizes the residual sum of squares,

$$\hat{\beta}^{\text{OLS}} = \arg\min_{\beta} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2, \tag{3}$$

with no penalty on coefficient magnitude, and serves as the baseline against which the regularized models are compared.

3.3 Ridge Regression

Ridge regression augments the least-squares objective with an ℓ_2 penalty on the coefficient vector:

$$\hat{\beta}^{\text{Ridge}} = \arg\min_{\beta} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2, \tag{4}$$

where $\lambda \geq 0$ controls the strength of shrinkage. As λ increases, coefficients are shrunk smoothly toward zero but, except in the limit, are never set exactly to zero.

3.4 LASSO

LASSO replaces the ℓ_2 penalty with an ℓ_1 penalty:

$$\hat{\beta}^{\text{LASSO}} = \arg\min_{\beta} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j|, \tag{5}$$

a formulation that admits corner solutions in coefficient space and, unlike Ridge, can set coefficients exactly to zero – performing simultaneous shrinkage and variable selection.

3.5 Elastic Net

The Elastic Net nests both penalties through a mixing parameter $\alpha \in [0, 1]$:

$$\hat{\beta}^{\text{EN}} = \arg\min_{\beta} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \left[\alpha \sum_{j=1}^{p} |\beta_j| + \frac{1-\alpha}{2} \sum_{j=1}^{p} \beta_j^2 \right], \tag{6}$$

recovering LASSO at $\alpha = 1$ and Ridge at $\alpha = 0$.

3.6 Estimation Procedure

The design matrix was constructed via one-hot encoding of all categorical predictors, with the intercept column removed prior to estimation since the regularized-regression routine includes an intercept by default. Ridge, LASSO, and Elastic Net were estimated using the coordinate-descent implementation in the `glmnet` package [2]. For Ridge and LASSO, the penalty parameter λ was selected by 10-fold cross-validation, choosing the value that minimizes the cross-validated mean squared error (`lambda.min`). For the Elastic Net, a grid search over $\alpha \in \{0, 0.1, 0.2, \ldots, 1.0\}$ was performed using a common fold assignment across all values of α, so that the comparison across mixing parameters is not confounded by different cross-validation splits; for each α, the minimum cross-validated error was recorded, and both α and λ were then chosen jointly to minimize this quantity. All random procedures used a fixed seed for reproducibility.

Finally, predictive performance was assessed out-of-sample using an 80/20 train–test split, with each of the four models re-fit on the training partition only and evaluated on the held-out test set via RMSE on the log-price scale.

4 Results

4.1 OLS Baseline

The OLS specification explains approximately 88.2% of the variance in log sale price (adjusted $R^2 = 0.8802$), with a residual standard error of 0.141 on 2,882 degrees of freedom ($F = 479.1$, $p < 2.2 \times 10^{-16}$). Table 1 reports the coefficients on the key structural and qualitative predictors.

Table 1: OLS coefficients on selected predictors (response: $\log$ SalePrice).

Predictor	Estimate	Std. Error	p-value
Intercept	5.4279	0.4282	< 0.001
Overall.Qual	0.0729	0.0035	< 0.001
Overall.Cond	0.0529	0.0028	< 0.001
Gr.Liv.Area	0.000232	0.0000118	< 0.001
Total.Bsmt.SF	0.000104	0.0000081	< 0.001
Garage.Cars	0.0547	0.0049	< 0.001
Year.Built	0.00268	0.00021	< 0.001
Full.Bath	0.0187	0.0074	0.012
TotRms.AbvGrd	−0.0015	0.0032	0.632
Lot.Area	0.00000195	0.00000039	< 0.001
Central.AirY	0.0969	0.0126	< 0.001

Overall quality and overall condition are both highly significant with positive coefficients, confirming that the market prices both the standard of materials and finish and the physical state of the property. `Kitchen.Qual` and `Central.Air` are also strongly significant. By contrast, `TotRms.AbvGrd` is not statistically distinguishable from zero once `Gr.Liv.Area` is included, since the raw room count adds little additional information beyond living area. Among the `Neighborhood` levels, several locations (Crawford, Northridge Heights, Stone Brook, Green Hills) command a significant premium relative to the baseline category, while others (Iowa DOT and Rail Road, Old Town, Briardale, Meadow Village) are priced significantly below it.

Figure 2 presents the standard OLS diagnostic panel. The residuals-versus-fitted and scale-location plots show broadly constant variance across the bulk of fitted values, with a small number of influential low-price outliers (observations 182, 1499, and 2181) visible in the Q–Q plot and in the Cook's distance panel; none, however, exceeds a Cook's distance of 1, so no observation was removed prior to the regularized-model comparison.

4.2 Ridge Regression

Cross-validation selected $\lambda^{\text{Ridge}} = 0.0336$ (Figure 3). As expected by construction, Ridge retains all predictors in the model, with zero coefficients set to exactly zero, confirming that the variable elimination observed for LASSO (below) is a genuine feature of the ℓ_1 penalty rather than an artifact of the cross-validation procedure itself.

4.3 LASSO

Cross-validation selected a considerably smaller penalty, $\lambda^{\text{LASSO}}_{\text{min}} = 0.000602$, reflecting the relatively flat, low-error region of the cross-validation curve once approximately 24–40 predictors are retained (Figure 4).

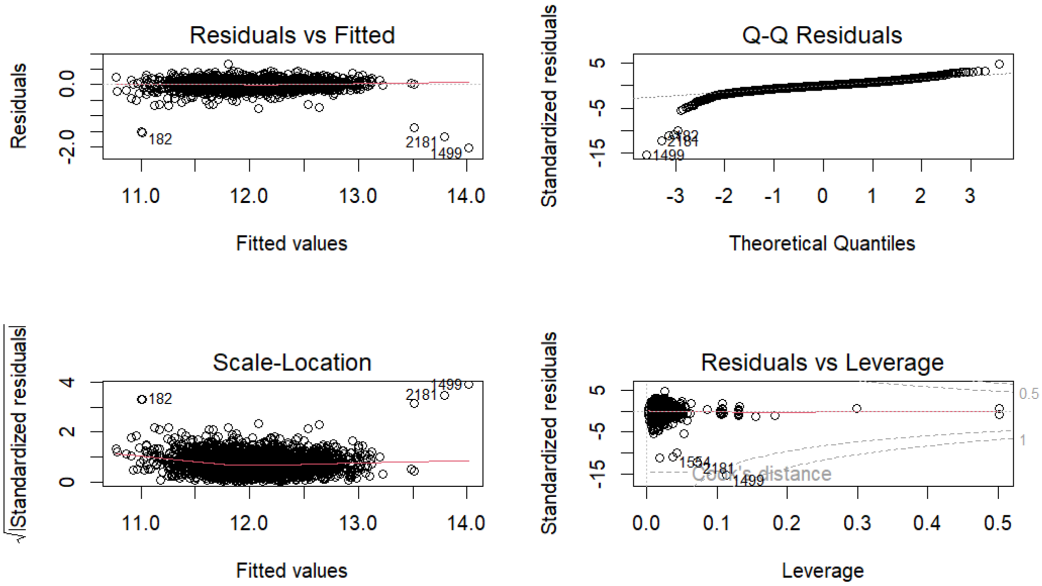

Figure 2: OLS diagnostic plots: residuals vs. fitted values, normal Q–Q plot, scale-location plot, and residuals vs. leverage with Cook's distance contours.

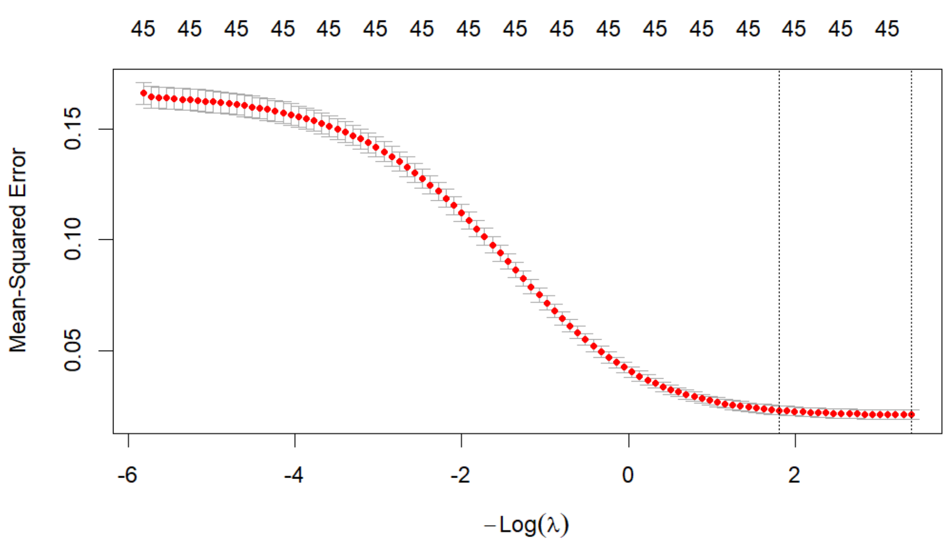

Figure 3: Ridge cross-validation error as a function of the penalty parameter. The number of non-zero coefficients (45, the full predictor set) is marked along the top axis and does not change with λ.

At this value, LASSO sets seven coefficients exactly to zero: `TotRms.AbvGrd` and the six weakest `Neighborhood` indicators (Gilbert, Landmark, North Ames, Sawyer, South & West of Iowa State University, and the `Kitchen.Qual = Po` level). These are precisely the categories that were also statistically weak or non-significant under OLS, indicating that the eliminated variables are those whose effects are not distinguishable from the reference category rather than economically important omissions.

Using the more conservative one-standard-error rule ($\lambda_{1se}^{\text{LASSO}}$), a substantially larger number of coefficients – 22, including `Full.Bath` and ten additional `Neighborhood` levels – are shrunk to zero, illustrating the trade-off between predictive optimality (`lambda.min`) and model parsimony (`lambda.1se`).

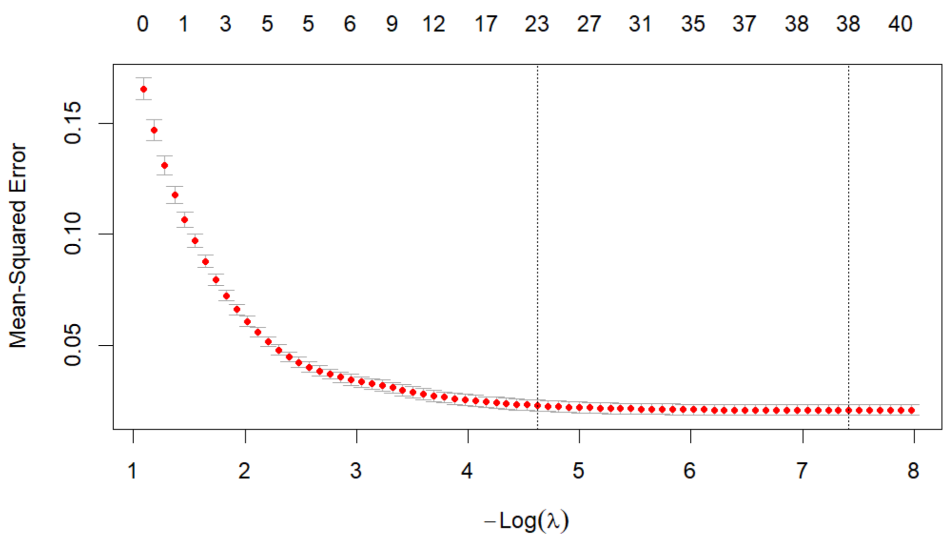

Figure 4: LASSO cross-validation error as a function of the penalty parameter, with the number of non-zero coefficients marked along the top axis.

The strongest predictors agree closely between OLS and LASSO: `Overall.Qual`, `Gr.Liv.Area`, `Year.Built`, `Total.Bsmt.SF`, `Garage.Cars`, and `Kitchen.Qual` retain large, stable coefficients under both estimators, supporting the interpretation that these are genuine drivers of price rather than artifacts of overfitting in the unregularized model. This is confirmed by the coefficient path plot in Figure 5, which traces each coefficient as a function of $-\log\lambda$: the paths that remain far from zero across a wide range of λ correspond to these core structural predictors, while paths that collapse to zero quickly correspond to the weak `Neighborhood` dummies identified above.

4.4 Elastic Net

The grid search over $\alpha \in \{0, 0.1, \ldots, 1.0\}$ produced the cross-validated errors reported in Table 2. The minimum cross-validated error is achieved at $\alpha = 0.1$, close to the Ridge end of the spectrum but with a small ℓ_1 component. The error surface is, however, fairly flat across α, with the ten-fold range of minimum errors spanning only 0.02075 to 0.02092.

At the selected $\alpha = 0.1$, the Elastic Net eliminates five predictors (Neighborhood: Gilbert, Landmark, Sawyer, SWISU; Kitchen.Qual: Po) – fewer than LASSO at `lambda.min`, consistent with the partial Ridge-like shrinkage induced by the small ℓ_1 weight.

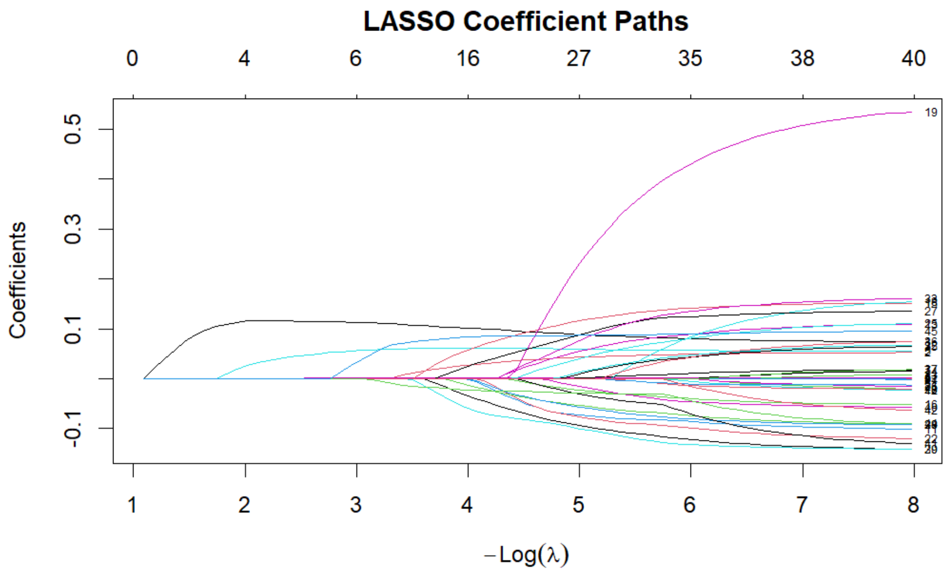

Figure 5: LASSO coefficient paths as a function of $-\log \lambda$. Each line traces one coefficient's trajectory from zero (at high λ) to its unpenalized value (at low λ).

Table 2: Cross-validated error by Elastic Net mixing parameter α.

α	0.0	0.1	0.2	0.5	0.8	1.0
CV error	0.02092	0.02075	0.02075	0.02076	0.02076	0.02076

4.5 Coefficient Comparison Across Models

Table 3 compares the coefficients of six key numeric predictors across all four estimators. The four models agree closely on both the sign and the approximate magnitude of these coefficients, indicating that the OLS estimates are not artifacts of overfitting: the regularized variants recover essentially the same relationships between predictors and log sale price.

Table 3: Coefficient comparison across models for selected numeric predictors (log-price scale).

Predictor	OLS	Ridge	LASSO	Elastic Net
Overall.Qual	0.0729	0.0707	0.0748	0.0740
Overall.Cond	0.0529	0.0460	0.0520	0.0513
Garage.Cars	0.0547	0.0590	0.0554	0.0558
Year.Built	0.00268	0.00253	0.00275	0.00271
Full.Bath	0.0187	0.0313	0.0171	0.0198
Lot.Area	0.00000195	0.00000180	0.00000200	0.00000198

Figure 6 presents the same comparison graphically. The bars for each predictor sit close together across the four models, visually confirming that regularization primarily rescales rather than reverses the estimated relationships; the one partial exception is `Full.Bath`, whose coefficient is noticeably higher under Ridge than under the other three estimators, consistent with Ridge's tendency to redistribute weight among mutually correlated predictors rather than to shrink any single one aggressively.

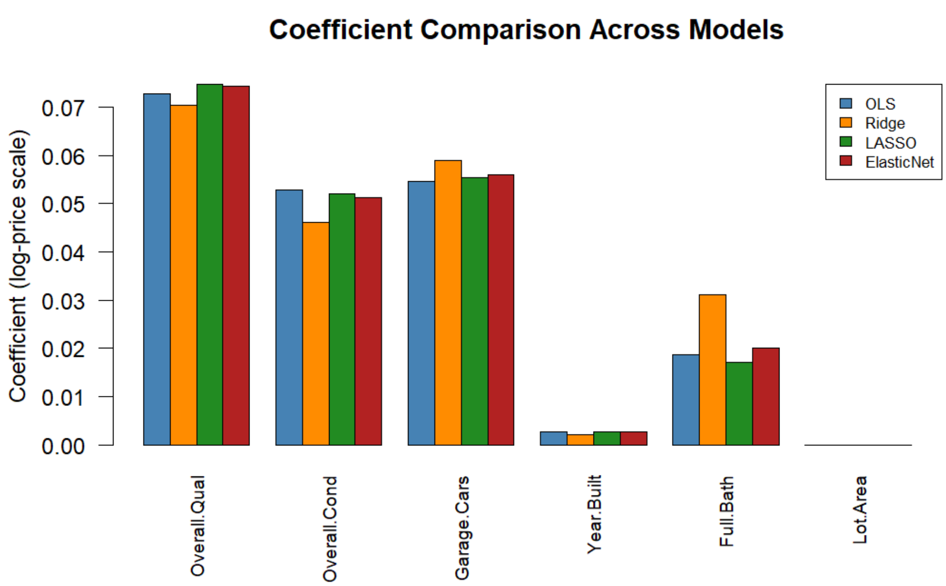

Figure 6: Coefficient comparison across the four models for six key numeric predictors.

4.6 Out-of-Sample Predictive Performance

Table 4 reports test-set RMSE on the log-price scale from an 80/20 train–test split, with each model independently re-fit on the training partition. The four models perform almost identically out-of-sample, with differences in RMSE appearing only in the third decimal place; Ridge attains the lowest test RMSE, marginally ahead of the Elastic Net, OLS, and LASSO. Figure 7 presents the same comparison graphically.

Table 4: Test-set RMSE by model (log-price scale).

Model	Test RMSE
OLS	0.16423
Ridge	0.16368
LASSO	0.16452
Elastic Net	0.16409

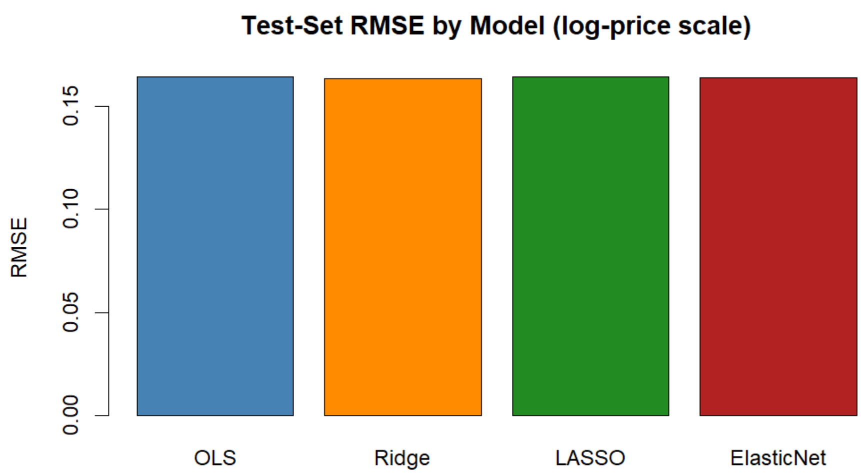

Figure 7: Test-set RMSE by model (log-price scale). The four bars are visually indistinguishable at this scale, underscoring how similar out-of-sample accuracy is across estimators.

The OLS predicted-versus-actual plot on the held-out test set (Figure 8) shows points tightly clustered around the 45-degree line across the bulk of the price distribution, with a small number of larger deviations at the extremes, consistent with the reported test RMSE and the in-sample R^2.

5 Discussion

Three findings stand out. First, all four estimators explain a very similar and substantial share of the variance in log sale price, both in-sample ($R^2 \approx 0.88$ for OLS) and out-of-sample (RMSE within roughly 0.0008 of one another). With $n = 2{,}928$ observations and only around 40 predictors after dummy expansion, the ratio of observations to parameters is large enough that OLS does not overfit severely; consequently, regularization does not translate into a measurable gain in predictive accuracy in this setting.

Second, the practical value of LASSO and the Elastic Net here lies in model parsimony and interpretability rather than in raw predictive performance. Both methods correctly identify and remove the same set of weak categorical levels – small, thinly populated `Neighborhood` categories whose effect cannot be distinguished from the baseline – along with the redundant `TotRms.AbvGrd` predictor, which is subsumed by `Gr.Liv.Area`. This is a useful contrast to emphasize in applied work: regularization is not always about

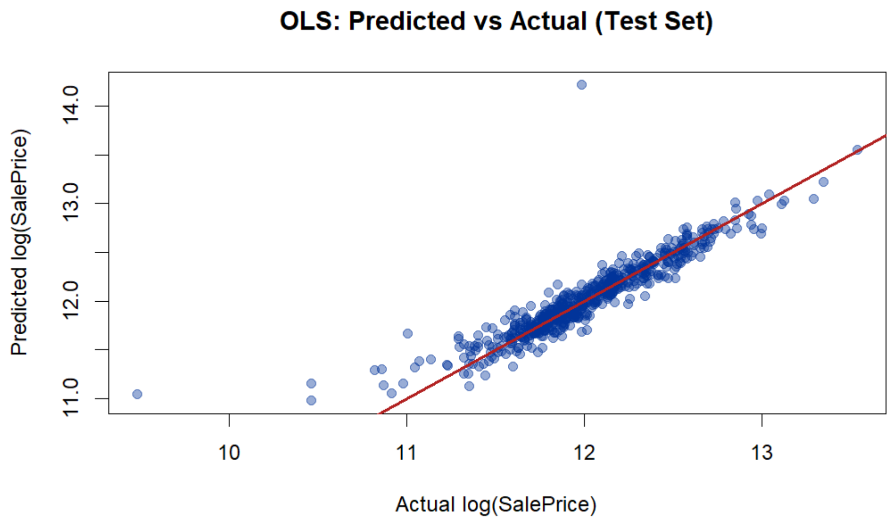

Figure 8: Predicted versus actual log sale price, OLS model, test set. The red line is the 45-degree reference.

beating OLS on prediction; its value can instead lie in producing a more parsimonious model at essentially no cost in accuracy.

Third, the choice between `lambda.min` and `lambda.1se` materially affects the degree of sparsity obtained from LASSO (seven versus twenty-two eliminated predictors), underscoring that variable-selection results from penalized regression are sensitive to the criterion used to select the tuning parameter, and should be reported alongside that choice.

A limitation of the present analysis is that only a fixed 80/20 split was used to assess out-of-sample performance; a more robust comparison would repeat the train–test evaluation across multiple random splits, or use nested cross-validation, to obtain a distribution of RMSE values for each model rather than a single point estimate. In addition, the log-linear specification, while appropriate for satisfying homoscedasticity, does not capture potential non-linearities or interaction effects (for example, between overall quality and living area), which could be explored using tree-based or spline-based extensions in future work.

6 Conclusion

This paper compared OLS, Ridge, LASSO, and Elastic Net regression for predicting log sale prices in the Ames Housing dataset. All four models achieve comparable in-sample fit ($R^2 \approx 0.88$) and near-identical out-of-sample RMSE, indicating that regularization offers little predictive advantage when the sample size is large relative to the number of predictors. The principal benefit of LASSO and the Elastic Net in this application is variable selection: both methods consistently identify and remove a small set of weak neighborhood categories and a redundant room-count predictor, while preserving the sign and approximate magnitude of the coefficients on the dataset's strongest and most economically meaningful predictors – overall quality, living area, basement area, garage capacity, and year built.

References

[1] De Cock, D. (2011). Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project. *Journal of Statistics Education*, 19(3).

[2] Friedman, J., Hastie, T., & Tibshirani, R. (2010). Regularization Paths for Generalized Linear Models via Coordinate Descent. *Journal of Statistical Software*, 33(1), 1–22.

[3] Hoerl, A. E., & Kennard, R. W. (1970). Ridge Regression: Biased Estimation for Nonorthogonal Problems. *Technometrics*, 12(1), 55–67.

[4] Tibshirani, R. (1996). Regression Shrinkage and Selection via the Lasso. *Journal of the Royal Statistical Society: Series B*, 58(1), 267–288.

[5] Zou, H., & Hastie, T. (2005). Regularization and Variable Selection via the Elastic Net. *Journal of the Royal Statistical Society: Series B*, 67(2), 301–320.